

It's About Time... and Neutrinos: Synchronizing the Pacific with P-ONE

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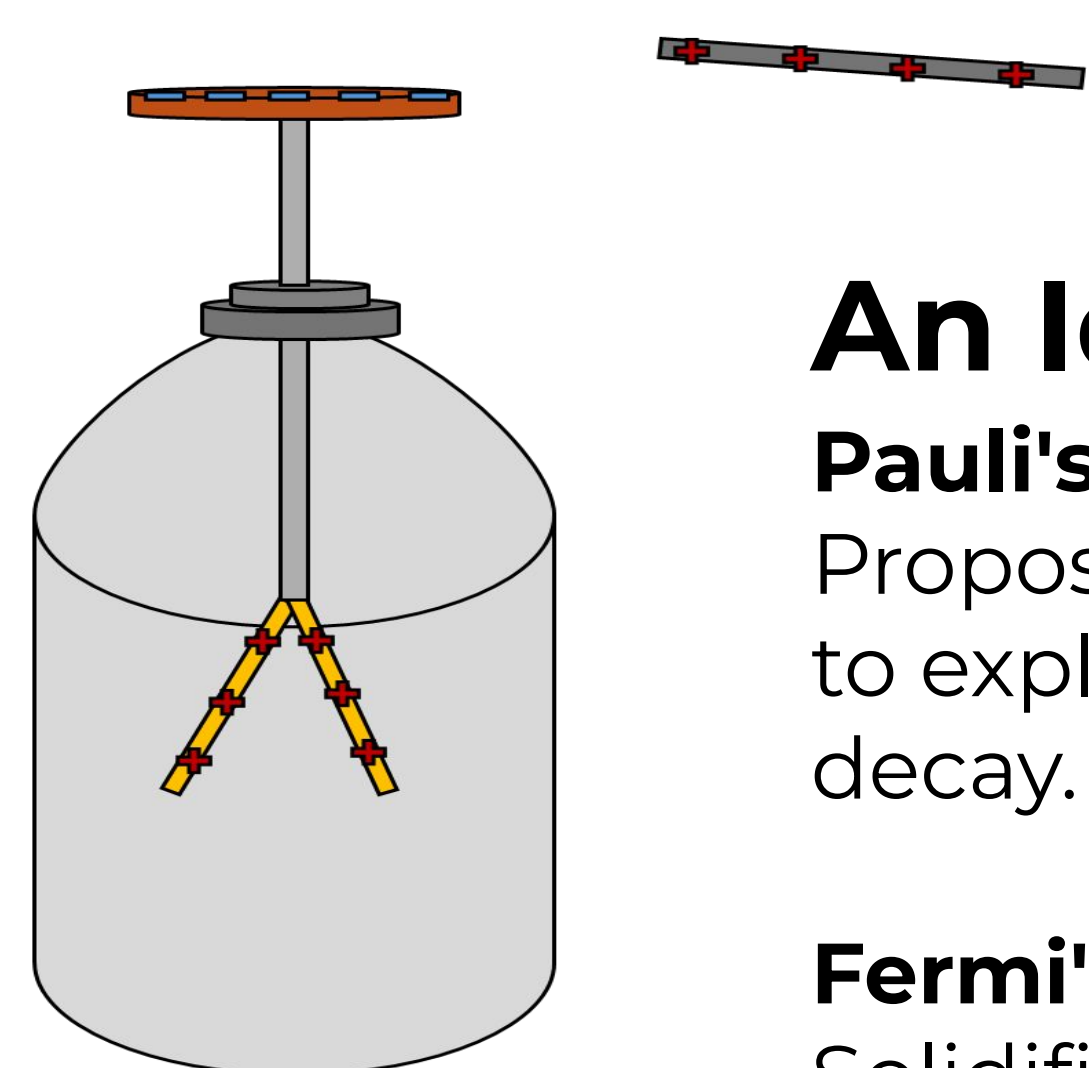
Abstract

The Pacific Ocean Neutrino Experiment (P-ONE) is an international collaboration advancing neutrino astrophysics, building on the pioneering IceCube project. Neutrinos — ghostly, elusive particles — probe distant galaxies and reveal cosmic particle accelerators through light deposition and timing patterns from their interactions. Enhancing timing accuracy improves our ability to trace these astrophysical sources. Our project centers on the timing system for the P-ONE-1 detector string. The Meinberg Syncbox locks to a global time base, generating Pulse-Per-Second and 10MHz signals for redundant timing systems. Clock signals are distributed through a custom timing board, with serial communications handling testing and calibration. This will significantly enhance neutrino detection, advancing astroparticle physics.

From Historical 'Wait, What?' to Cosmic 'Zoom Zoom'

Coulomb's Electroscope

Early Curiosity (1785): Coulomb's electroscope demonstrated unexpected discharging phenomena.



An Idea Is Born

Pauli's Hypothesis (1930)

Proposed a neutral, light particle to explain missing energy in beta decay.

Fermi's Theory (1934)

Solidified the concept of beta decay.

Sources and Cosmic Accelerators

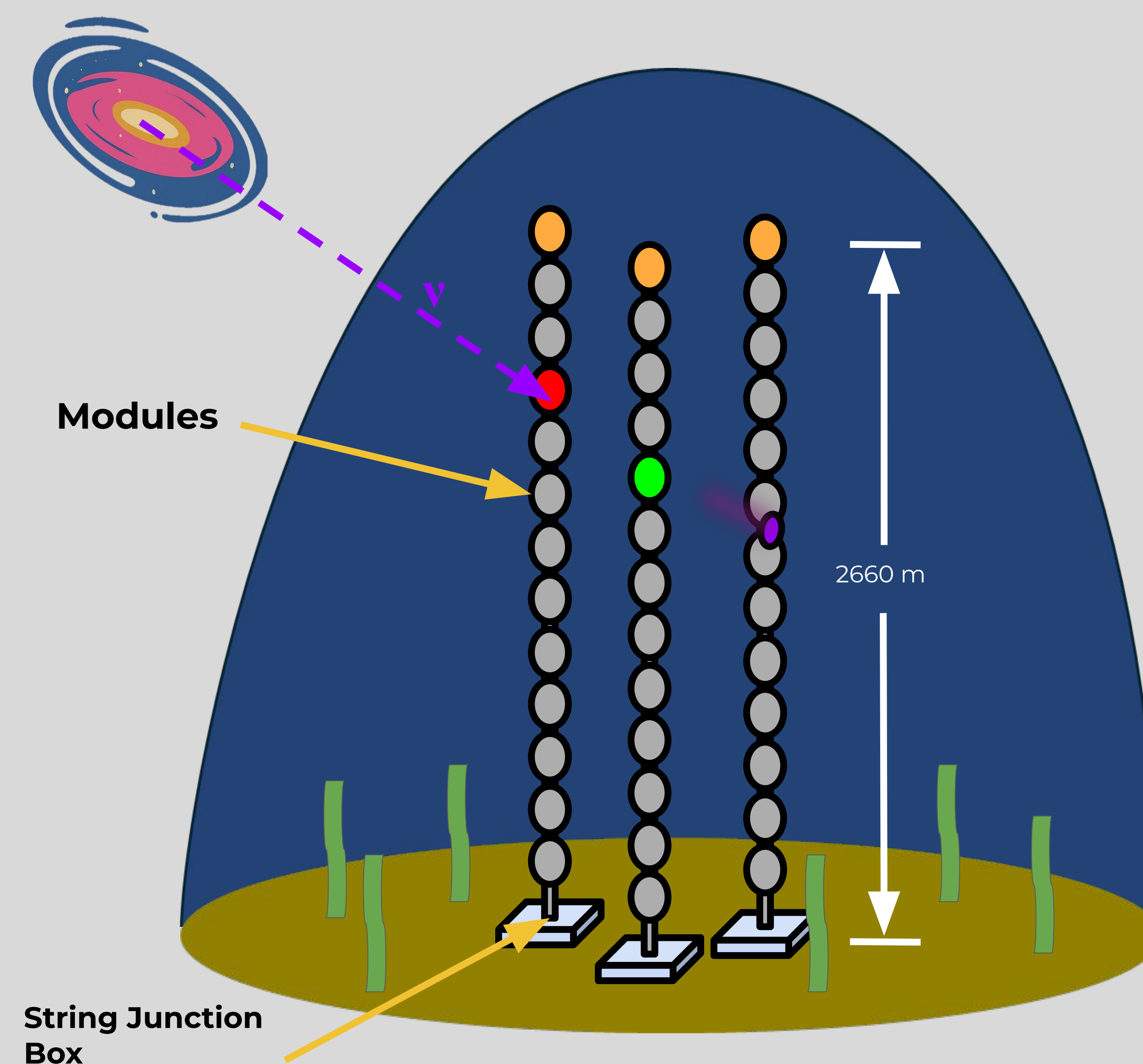
Natural v.s. Artificial

- Neutrinos come from many sources
- Not all neutrinos are the same



Cosmic accelerators: Some neutrinos are produced in cosmic events like supernovae, gamma-ray bursts, and the interactions of cosmic rays with interstellar matter. These high-energy environments accelerate particles to nearly the speed of light, creating neutrinos.

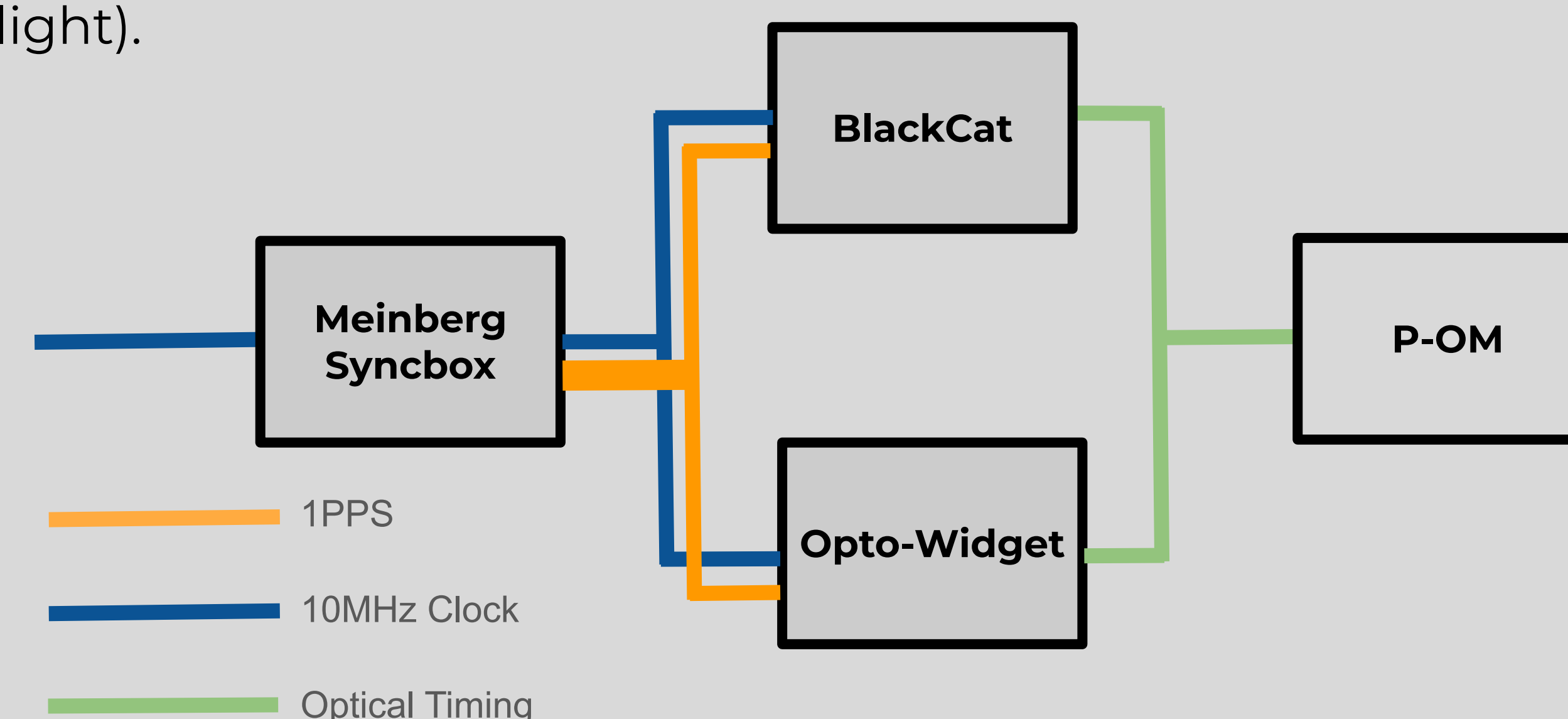
Pacific Ocean Neutrino Experiment



The current plan is to deploy the P-ONE-1 by August 2025 within the Victorian Basin. (Unfortunately, there's a restriction on having us "nerds" on the boat → sad face)

By the end of the project, we will have **70 strings with 1,400 optical modules** to aid in reconstructing the linear trajectories of secondaries induced by neutrinos. These will all be synchronized to a handful of picoseconds and connected over gigabit fiber ethernet connections – our optical modules could watch Netflix in the depth of the ocean.

Since neutrinos themselves cannot be directly detected, we instead observe the secondary particles produced from their interactions, which generate Cherenkov radiation (light blue light).



Theory of operations

BlackCat: Has the ability to distribute time (~ 50 ps) with the addition of data transfer (GB/s)

Opto-Widget: Disperses time (~ 10 ps)

Meinberg clock: Synchronizes time into the system from the eternal network

Methodology

Before engaging with the Meinberg Syncbox and Opto-Widget, we first conducted preliminary work with the White Rabbit Nodes and Zedboard.

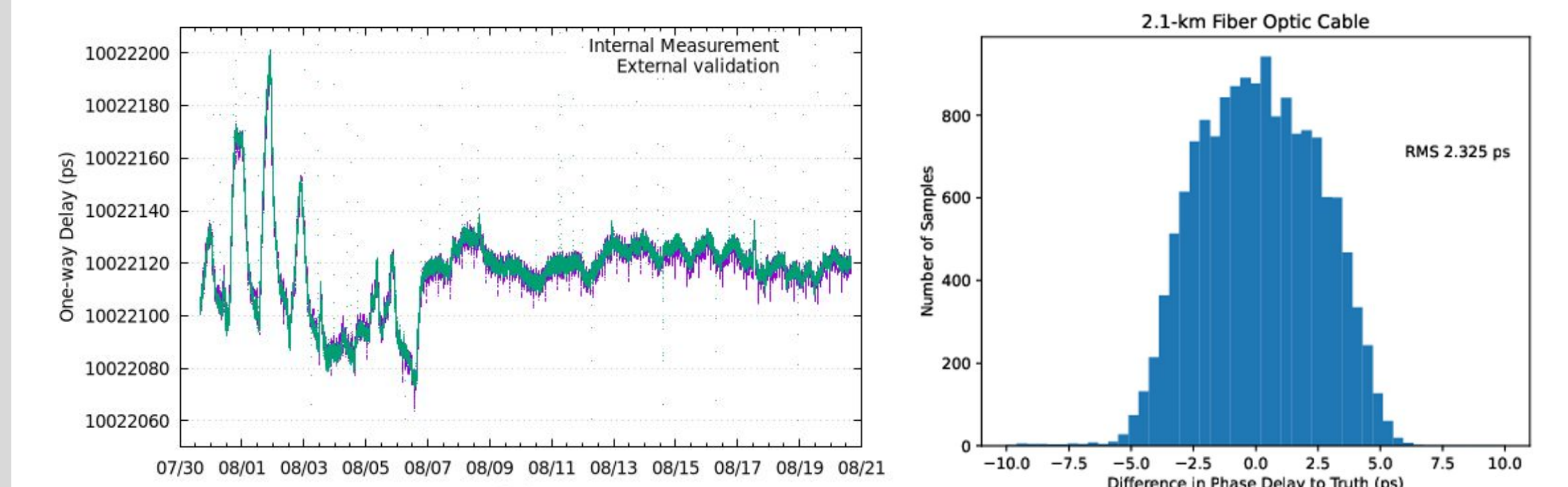
Upon receiving the Meinberg Syncbox, we leveraged Prof. Froehlich's expert guidance and utilized multiple power supplies to power the device.

Serial Communications: It is essential to establish proper communication between the Meinberg Syncbox and the Opto-Widget, ensuring reliable serial connections between all components.

Timing Distribution: Fiber optic cables carry "clock" signals at 10MHz modulated with the PPS time-of-day or "drumbeat" signal. The modulation is unraveled by a timing chip commonly used in 5G cell towers on each end allowing picosecond synchronization of the systems clocks to each other and microsecond synchronization to global time.

Results

To validate the performance of our timing system, we've completed a clock synchronization and round trip delay measurement. The system built here at Elmhurst University was able to synchronize times between our two custom printed circuit boards by precisely measuring the time-of-flight of fiber optic signals between them, and then synchronizing clocks using the computed delays. This method allows us to synchronize signals on the two boards to within 2.2 ps (2.2 trillionths of a second, or 2.2×10^{-12} s), thereby allowing the recording of the neutrino-induced light amongst optical modules to be precisely synchronized.



This level of synchronization will allow the full 70 string P-ONE detector to achieve a reconstruction precision of less than .05° on average, enabling quick discovery of new sources. Additionally, the P-ONE telescope will view a complementary portion of the sky to the successful IceCube telescope.

Acknowledgements

A heartfelt thanks to all the P-ONE collaborators for your invaluable support and dedication. Your contributions have been crucial to the success of this project. We are grateful for your teamwork and commitment.

We extend our thanks to the Physics Department for their support. Special thanks to Dr. Halliday and Bob Froehlich for their invaluable guidance. We also appreciate the CASE program and HIP grant, which were crucial to the success of this project.

